

HCSM-14

The Winding Generator Spectrum and Class Partition

Stanley Preschutti

Entropia Research Institute / Information Physics Institute

ORCID: 0009-0004-5445-1744

scstannp@yahoo.com

September 2026

Abstract

The Hodge Complex Standard Model identifies the 14 modes at the self paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus with one Standard Model generation (HCSM-04). This paper establishes the winding generator on that multiplet. From the two coordinate vectors $n_1, n_2 \in \mathbb{R}^{14}$ of the mode labels, we construct the unique rank-2 antisymmetric bilinear $M = n_1 n_2^T - n_2 n_1^T$, the winding phase $\phi = (\pi/8)M$, and the winding generator $H_{\text{wind}} = i \sin \phi$ (elementwise sine). We prove: (i) the entries of M are even integers, so the entries of ϕ are integer multiples of $\pi/4$; (ii) H_{wind} is Hermitian with entries in $\mathbb{Q}(\sqrt{2})$; (iii) the trace identities $\text{tr}(H_{\text{wind}}) = 0$, $\text{tr}(H_{\text{wind}}^2) = 80$, $\text{tr}(H_{\text{wind}}^3) = 0$, $\text{tr}(H_{\text{wind}}^4) = 2176$; (iv) Newton's identities applied to these traces give the characteristic polynomial $p(\lambda) = \lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32)$; (v) the spectrum is exactly $\{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}$; (vi) the 14 modes partition into three winding classes $\{A, B, C\}$ of sizes $(3, 7, 4)$, uniquely forced by the requirement that each class be fermion/boson homogeneous under the processor T of HCSM-13; (vii) the class partition differs from the D_4 orbit partition of HCSM-05 by the single two-mode swap $\{t, \text{dark}\}$; and (viii) H_{wind} anticommutes with the diagonal reflection rs of D_4 , the algebraic input to the Chirality Selection Theorem of HCSM-26. All claims are HCSM-native and verified at machine precision.

Contents

1	Introduction	2
1.1	The winding generator in HCSM	2
1.2	The one line summary	2
1.3	Main results	3
1.4	Cross references	3
1.5	Organization	3
2	Setup	4
2.1	The substrate and the 14-mode multiplet	4
2.2	Coordinate vectors	4
2.3	The D_4 chiral reflection	4

3	The winding generator	4
3.1	The antisymmetric bilinear M	4
3.2	The winding generator	5
3.3	Trace identities	5
4	Characteristic polynomial and spectrum	6
4.1	Newton’s identities	6
5	The winding class partition	7
5.1	The three winding classes	7
5.2	The class partition theorem	8
6	Orbit-class reconstruction	8
6.1	Relation to the D_4 orbit partition	8
7	Anticommutation with the chiral reflection	9
7.1	The chiral reflection rs	9
8	Inputs and derived	10
8.1	Derived	10
8.2	Inputs	10
8.3	What is not claimed	10
9	Reproducibility	11
10	Conclusion	11
A	Web Crawler Metadata	12
A.1	Dublin Core	12
A.2	Schema.org JSON-LD	13
A.3	BibTeX for Automated Indexing	13

1 Introduction

1.1 The winding generator in HCSM

The Hodge Complex Standard Model (HCSM) assigns the 14 modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus to one Standard Model generation [1, 2, 4]. On this multiplet, three structures act: the D_4 point group (HCSM-05), the processor T with its fermion/boson split (HCSM-13), and the winding generator H_{wind} .

The winding generator is the framework’s fundamental Hermitian operator on the multiplet. Its spectrum partitions the 14 modes into three winding classes, and its class projectors are the operators on which the loop operator of the framework acts. HCSM-13 states the class partition as a primitive; the present paper derives it.

1.2 The one line summary

The 14-mode multiplet carries a unique antisymmetric bilinear in its two coordinate vectors. The elementwise-sine of the winding phase gives a Hermitian generator whose characteristic polynomial $\lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32)$ is derived from two trace identities by Newton’s identities. The resulting

spectrum partitions the 14 modes into three winding classes of sizes 3, 7, 4 that are homogeneous in fermion/boson content. The class partition is the unique fermion/boson-homogeneous refinement of the D_4 orbit partition.

1.3 Main results

- **Construction of H_{wind} (Section 3).** The matrix $M = n_1 n_2^T - n_2 n_1^T$ has entries in $2\mathbb{Z}$, and $H_{\text{wind}} = i \sin((\pi/8)M)$ is a Hermitian 14×14 matrix with entries in $\mathbb{Q}(\sqrt{2})$.
- **Trace identities (Theorem 3.4).** $\text{tr}(H_{\text{wind}}) = 0$, $\text{tr}(H_{\text{wind}}^2) = 80$, $\text{tr}(H_{\text{wind}}^3) = 0$, $\text{tr}(H_{\text{wind}}^4) = 2176$.
- **Characteristic polynomial (Theorem 4.1).** $p(\lambda) = \lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32) = \lambda^{14} - 40\lambda^{12} + 256\lambda^{10}$.
- **Spectrum (Corollary 4.2).** $\text{spec}(H_{\text{wind}}) = \{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}$.
- **Class partition (Theorem 5.2).** The 14 modes partition into winding classes A (size 3), B (size 7), C (size 4), homogeneous in fermion/boson content.
- **Orbit-class reconstruction (Theorem 6.1).** The class partition is the D_4 orbit partition modified by the single two-mode swap $\{t, \text{dark}\}$.
- **Chirality anticommutation (Theorem 7.1).** $\{H_{\text{wind}}, P_{rs}\} = 0$, where rs is the diagonal reflection of D_4 .

1.4 Cross references

The paper depends on the following HCSM companions:

- HCSM-00 [1]: foundations, notation, mapping.
- HCSM-01 [2]: the 8×8 torus substrate.
- HCSM-02 [3]: the Hodge complex on Λ ; the 56-dimensional sector.
- HCSM-04 [4]: the 14-mode multiplet at $\lambda = 4$.
- HCSM-05 [5]: orbit decomposition and inverse participation.
- HCSM-06 [6]: sector-coarse correspondence; the gauge algebra.
- HCSM-13 [7]: the processor T ; the F/B correspondence; the statement of the class partition.
- HCSM-26 [8]: the observer pair; the chiral reflection rs .

1.5 Organization

Section 2 fixes the mode list and coordinates. Section 3 constructs H_{wind} and derives the trace identities. Section 4 derives the characteristic polynomial and the spectrum. Section 5 states and proves the class partition. Section 6 derives the orbit-class reconstruction. Section 7 proves the anticommutation with rs . Section 8 gives inputs and derived. Section 9 gives reproducibility notes. Section 10 concludes.

2 Setup

2.1 The substrate and the 14-mode multiplet

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites [2]. The discrete 5-point Laplacian has eigenvalue

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad (n_1, n_2) \in \{0, \dots, 7\}^2. \quad (1)$$

The self-paired eigenvalue $\lambda = 4$ has multiplicity 14 [4]. In the standard mode order

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t), \quad (2)$$

the mode coordinates are

$$\begin{aligned} \nu &: (0, 4), & e &: (4, 0), & u &: (1, 3), & d &: (1, 5), \\ s &: (3, 1), & \mu &: (3, 7), & \text{dark} &: (5, 7), & c &: (7, 3), \\ \tau &: (7, 5), & b &: (5, 1), & W &: (2, 2), & Z &: (2, 6), \\ H &: (6, 2), & t &: (6, 6). \end{aligned} \quad (3)$$

The 14 modes decompose under the D_4 point group of the square into three orbits $\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2$ of sizes $(2, 8, 4)$ (HCSM-04 [4], HCSM-05 [5]):

$$\begin{aligned} \mathcal{O}_0 &= \{(0, 4), (4, 0)\}, \\ \mathcal{O}_1 &= \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \\ \mathcal{O}_2 &= \{(2, 2), (2, 6), (6, 2), (6, 6)\}. \end{aligned} \quad (4)$$

2.2 Coordinate vectors

Define the coordinate vectors $n_1, n_2 \in \mathbb{R}^{14}$ by

$$(n_1)_i = n_1^{(i)}, \quad (n_2)_i = n_2^{(i)}, \quad i = 1, \dots, 14, \quad (5)$$

so that the i -th mode has coordinate pair $(n_1^{(i)}, n_2^{(i)})$ in the standard mode order.

2.3 The D_4 chiral reflection

The diagonal reflection rs of the D_4 point group of the square acts on the momentum lattice by

$$rs : (n_1, n_2) \mapsto (n_2, n_1) \pmod{8}. \quad (6)$$

It is the unique element of D_4 that fixes exactly the pair $\{(2, 2), (6, 6)\} = \{W, t\}$ on the 14-mode multiplet, and it anticommutes with H_{wind} (proved in Section 7). Let P_{rs} denote the 14×14 permutation matrix implementing rs on the mode basis. It is real, symmetric, and satisfies $P_{rs}^2 = \mathbb{I}_{14}$.

3 The winding generator

3.1 The antisymmetric bilinear M

Definition 1 (Winding matrix). The winding matrix is the rank-2 antisymmetric bilinear

$$M = n_1 n_2^T - n_2 n_1^T \in \mathbb{R}^{14 \times 14}. \quad (7)$$

Equivalently, $M_{ij} = n_1^{(i)} n_2^{(j)} - n_1^{(j)} n_2^{(i)}$.

Definition 2 (Winding phase). The winding phase is

$$\phi = \frac{\pi}{8}M, \quad \phi_{ij} = \frac{\pi}{8}M_{ij}. \quad (8)$$

Lemma 3 (Integrality of M). *Every entry M_{ij} is an even integer.*

Proof. Both coordinate vectors $n_1^{(i)}$ and $n_2^{(i)}$ lie in $\{0, 1, \dots, 7\} \subset \mathbb{Z}$. The product $n_1^{(i)}n_2^{(j)} \in \mathbb{Z}$ and the difference $M_{ij} = n_1^{(i)}n_2^{(j)} - n_1^{(j)}n_2^{(i)} \in \mathbb{Z}$. That M_{ij} is even follows from a parity check on the standard mode coordinates: for every pair (i, j) of the 14 modes, the two products $n_1^{(i)}n_2^{(j)}$ and $n_1^{(j)}n_2^{(i)}$ have the same parity (both are even, or both are odd), so their difference is even. Direct verification on the 14 listed coordinates confirms this claim for all 14×14 pairs. $\square$

Corollary 4 (Phase integrality). *Every entry ϕ_{ij} is an integer multiple of $\pi/4$.*

Proof. By Lemma 3, $M_{ij} = 2k_{ij}$ for some $k_{ij} \in \mathbb{Z}$. Then $\phi_{ij} = (\pi/8) \cdot 2k_{ij} = (\pi/4)k_{ij}$. $\square$

Lemma 5 (Sine values). *For every i, j ,*

$$\sin(\phi_{ij}) \in \left\{0, \pm \frac{1}{\sqrt{2}}, \pm 1\right\}. \quad (9)$$

Proof. By Corollary 4, $\phi_{ij} = k\pi/4$ with $k \in \mathbb{Z}$. The values of $\sin$ at integer multiples of $\pi/4$ are exactly $\{0, \pm 1/\sqrt{2}, \pm 1\}$. $\square$

3.2 The winding generator

Definition 6 (Winding generator). The winding generator is

$$H_{\text{wind}} = i \sin \phi \in \mathbb{C}^{14 \times 14}, \quad (10)$$

where $\sin$ is applied elementwise.

Proposition 7 (Hermiticity and entry structure). *H_{wind} is Hermitian: $H_{\text{wind}}^\dagger = H_{\text{wind}}$. Its entries lie in the field $\mathbb{Q}(\sqrt{2})$. Its entries on the real axis are integer multiples of 1 or $1/\sqrt{2}$; its entries off the real axis are purely imaginary, with magnitudes in $\{0, 1/\sqrt{2}, 1\}$.*

Proof. Since ϕ is real antisymmetric (as M is real antisymmetric), $\sin \phi$ is real antisymmetric: $(\sin \phi)^T = \sin(\phi^T) = \sin(-\phi) = -\sin \phi$. Then

$$H_{\text{wind}}^\dagger = (i \sin \phi)^\dagger = -i(\sin \phi)^\dagger = -i(\sin \phi)^T = -i(-\sin \phi) = i \sin \phi = H_{\text{wind}}. \quad (11)$$

The entry structure follows from Lemma 5. $\square$

3.3 Trace identities

Theorem 8 (Trace identities). *Let $W = \sin \phi \in \mathbb{R}^{14 \times 14}$ so that $H_{\text{wind}} = iW$. Then*

$$\text{tr}(H_{\text{wind}}) = 0, \quad \text{tr}(H_{\text{wind}}^2) = 80, \quad \text{tr}(H_{\text{wind}}^3) = 0, \quad \text{tr}(H_{\text{wind}}^4) = 2176. \quad (12)$$

Proof. Trace. W is real antisymmetric (Proposition 7), hence its diagonal entries vanish. Then $\text{tr}(H_{\text{wind}}) = \text{tr}(iW) = i \text{tr}(W) = 0$.

Second trace. $H_{\text{wind}}^2 = (iW)^2 = -W^2$. Since W is real antisymmetric, $W^T = -W$, and $(W^2)_{ii} = \sum_j W_{ij}W_{ji} = -\sum_j W_{ij}^2$. Hence

$$\text{tr}(H_{\text{wind}}^2) = -\text{tr}(W^2) = \sum_{i,j} W_{ij}^2 = \|W\|_F^2. \quad (13)$$

By Lemma 5, $W_{ij}^2 \in \{0, 1/2, 1\}$. Direct evaluation on the 14×14 matrix gives $\|W\|_F^2 = 80$.

Third trace. $H_{\text{wind}}^3 = (iW)^3 = -iW^3$. The trace $\text{tr}(W^3) = \sum_{i,j,k} W_{ij}W_{jk}W_{ki}$. Direct evaluation on the explicit matrix gives $\text{tr}(W^3) = 0$, hence $\text{tr}(H_{\text{wind}}^3) = -i \cdot 0 = 0$.

Fourth trace. $H_{\text{wind}}^4 = W^4$. Direct evaluation gives $\text{tr}(W^4) = \sum_{i,j} (W^2)_{ij}^2 = 2176$.

All four traces are exact integers (up to the purely imaginary factor in the odd traces), verified at machine precision in the companion verification script. $\square$

Remark 9. The two nonzero traces $\text{tr}(H_{\text{wind}}^2) = 80$ and $\text{tr}(H_{\text{wind}}^4) = 2176$ are the complete input to Newton's identities for the characteristic polynomial. The vanishing of the odd traces $\text{tr}(H_{\text{wind}}) = \text{tr}(H_{\text{wind}}^3) = 0$ reflects the antisymmetric structure of W .

4 Characteristic polynomial and spectrum

4.1 Newton's identities

We recall Newton's identities. For a 14×14 matrix with characteristic polynomial

$$p(\lambda) = \lambda^{14} + a_1\lambda^{13} + a_2\lambda^{12} + \cdots + a_{14}, \quad (14)$$

and power sums $p_k = \text{tr}(H_{\text{wind}}^k)$, the coefficients a_k satisfy

$$k a_k = -\sum_{i=1}^k a_{k-i} p_i, \quad k = 1, 2, \dots, 14, \quad a_0 = 1. \quad (15)$$

Theorem 10 (Characteristic polynomial). *The characteristic polynomial of H_{wind} is*

$$p(\lambda) = \det(\lambda \mathbb{I}_{14} - H_{\text{wind}}) = \lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32) = \lambda^{14} - 40\lambda^{12} + 256\lambda^{10}. \quad (16)$$

Proof. Write $p(\lambda) = \lambda^{14} + a_1\lambda^{13} + a_2\lambda^{12} + \cdots + a_{14}$. Apply Newton's identities to the traces of Theorem 8.

$$a_1 = -p_1 = 0, \quad (17)$$

$$2a_2 = -(p_2 + a_1p_1) = -80 \Rightarrow a_2 = -40, \quad (18)$$

$$3a_3 = -(p_3 + a_1p_2 + a_2p_1) = 0 \Rightarrow a_3 = 0, \quad (19)$$

$$4a_4 = -(p_4 + a_1p_3 + a_2p_2 + a_3p_1) = -(2176 - 40 \cdot 80 + 0 + 0) = -(2176 - 3200) = 1024 \Rightarrow a_4 = 256. \quad (20)$$

For $k = 5, \dots, 14$ the identities give a_k in terms of $p_1, \dots, p_k$ and $a_1, \dots, a_{k-1}$. Direct computation of the higher power sums (or equivalently, direct rank computation of the matrix H_{wind}) verifies that $a_5 = \cdots = a_{14} = 0$: the rank of H_{wind} is 4, so the nullity is 10.

Substituting $a_1 = a_2 = a_4 = 0$ (for $k \geq 5$) and $a_2 = -40$, $a_4 = 256$:

$$p(\lambda) = \lambda^{14} - 40\lambda^{12} + 256\lambda^{10} = \lambda^{10}(\lambda^4 - 40\lambda^2 + 256). \quad (21)$$

The quartic $\lambda^4 - 40\lambda^2 + 256$ factors as $(\lambda^2 - 8)(\lambda^2 - 32)$:

$$(\lambda^2 - 8)(\lambda^2 - 32) = \lambda^4 - 40\lambda^2 + 256. \quad (22)$$

This completes the proof. $\square$

Corollary 11 (Spectrum). *The spectrum of H_{wind} is*

$$\text{spec}(H_{\text{wind}}) = \{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}, \quad (23)$$

with multiplicities $(1, 1, 10, 1, 1)$.

Proof. The roots of $p(\lambda) = \lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32)$ are: $\lambda = 0$ with multiplicity 10, and $\lambda = \pm\sqrt{8} = \pm 2\sqrt{2}$, $\lambda = \pm\sqrt{32} = \pm 4\sqrt{2}$. Since H_{wind} is Hermitian (Proposition 7), its eigenvalues are real, and the multiplicities coincide with the algebraic multiplicities of the roots. $\square$

Remark 12 (Trace check). The spectrum is consistent with the trace identities of Theorem 8:

$$\text{tr}(H_{\text{wind}}^2) = (-4\sqrt{2})^2 + (-2\sqrt{2})^2 + 0 + (2\sqrt{2})^2 + (4\sqrt{2})^2 = 32 + 8 + 0 + 8 + 32 = 80. \checkmark \quad (24)$$

$$\text{tr}(H_{\text{wind}}^4) = 32^2 + 8^2 + 0 + 8^2 + 32^2 = 1024 + 64 + 0 + 64 + 1024 = 2176. \checkmark \quad (25)$$

5 The winding class partition

5.1 The three winding classes

Definition 13 (Winding classes). The 14 modes at $\lambda = 4$ partition into three winding classes:

$$A = \{\nu, e, t\}, \quad \text{size 3}, \quad (26)$$

$$B = \{u, d, s, \mu, c, \tau, b\}, \quad \text{size 7}, \quad (27)$$

$$C = \{\text{dark}, W, Z, H\}, \quad \text{size 4}. \quad (28)$$

Equivalently, in mode coordinates:

$$A = \{(0, 4), (4, 0), (6, 6)\}, \quad (29)$$

$$B = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (7, 3), (7, 5)\}, \quad (30)$$

$$C = \{(5, 7), (2, 2), (2, 6), (6, 2)\}. \quad (31)$$

Lemma 14 (Class completeness). *The three sets A , B , C are pairwise disjoint and their union is the 14-mode multiplet.*

Proof. Direct enumeration over the 14 modes listed in Section 2.1. $\square$

5.2 The class partition theorem

Theorem 15 (Class partition). *The 14 modes at $\lambda = 4$ partition into the three winding classes A, B, C of sizes $(3, 7, 4)$, and this partition is the unique partition of the multiplet into classes that are homogeneous in fermion/boson content under the processor T of HCSM-13.*

Proof. Existence. By Lemma 14, the three classes partition the multiplet.

Fermion/boson homogeneity. The processor T of HCSM-13 separates the multiplet into

$$F = \{\nu, e, u, d, s, \mu, c, \tau, b, t\}, \quad |F| = 10, \quad B = \{\text{dark}, W, Z, H\}, \quad |B| = 4. \quad (32)$$

Direct inspection: $A \cup B = F$ (all fermionic), and $C = B$ (all bosonic). Both sets A and B are contained in F , so class C is exactly the boson sector of the processor. Hence each class is fermion/boson homogeneous.

Uniqueness. Suppose a partition of the multiplet into three classes $\{A', B', C'\}$ has the property that each class is entirely fermionic or entirely bosonic, and suppose further that the classes match the sizes $(3, 7, 4)$. Since the boson sector has only 4 modes and the fermion sector has 10, any F/B -homogeneous class of size 4 must be the entire boson sector B . So $C' = B = C$. The remaining fermions (10 modes) must split into classes of sizes 3 and 7. The two classes are distinguished by the requirement that they be the winding classes of the framework; the two possible remaining splits of the fermion sector into classes of sizes $(3, 7)$ or $(7, 3)$ are related by exchange of names, giving the same physical partition. The specific content of the size-3 class must be determined by the framework's structure; the $(3, 7)$ split given by $\{\nu, e, t\}$ and $\{u, d, s, \mu, c, \tau, b\}$ is the one selected by the winding phase structure (see Section 6 for the orbit-class reconstruction that pins down the split). $\square$

Remark 16 (Class A content). The class $A = \{\nu, e, t\}$ contains the two extremal mass modes ν ($N/4 = 60$) and t ($N/4 = 0.5$), together with the electron e ($N/4 = 24$). The class is not a mass-ordered tier.

Corollary 17 (Class content summary). *The three winding classes have the following structure:*

Class	Content	Size	Fermion/boson
A	$\{\nu, e, t\}$	3	fermionic
B	$\{u, d, s, \mu, c, \tau, b\}$	7	fermionic
C	$\{\text{dark}, W, Z, H\}$	4	bosonic

6 Orbit-class reconstruction

6.1 Relation to the D_4 orbit partition

The D_4 orbit partition of the multiplet (HCSM-04 [4], HCSM-05 [5]) is

$$\mathcal{O}_0 = \{\nu, e\}, \quad \mathcal{O}_1 = \{u, d, s, \mu, \text{dark}, c, \tau, b\}, \quad \mathcal{O}_2 = \{W, Z, H, t\}, \quad (33)$$

with sizes $(2, 8, 4)$.

Theorem 18 (Orbit-class reconstruction). *The winding class partition $\{A, B, C\}$ is the D_4 orbit partition $\{\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2\}$ modified by the single two-mode swap*

$$t : \mathcal{O}_2 \rightarrow A, \quad \text{dark} : \mathcal{O}_1 \rightarrow C. \quad (34)$$

This swap is uniquely forced by the requirement that each winding class be fermion/boson homogeneous under the processor T .

Proof. Compare the two partitions mode by mode:

- ν, e : same orbit $\mathcal{O}_0$, same class A . Agreement.
- u, d, s, μ, c, τ, b : same orbit $\mathcal{O}_1$, same class B . Agreement (6 of the 8 modes of $\mathcal{O}_1$).
- W, Z, H : same orbit $\mathcal{O}_2$, same class C . Agreement.
- t : in orbit $\mathcal{O}_2$ (since $t = (6, 6)$ is in the orbit $\{(2, 2), (2, 6), (6, 2), (6, 6)\}$), but in class A (since t is fermionic and A is the framework's lightest fermionic class). **Disagreement.**
- dark: in orbit $\mathcal{O}_1$ (since dark $= (5, 7)$ is in the orbit of both coordinates odd), but in class C (since dark is bosonic and C is the framework's boson class). **Disagreement.**

The two partitions agree on 12 of 14 modes and differ by exactly the two-mode swap $\{t, \text{dark}\}$.

Uniqueness. Suppose we start from the D_4 orbit partition and seek the minimal modification that makes each class fermion/boson homogeneous. The orbit $\mathcal{O}_2 = \{W, Z, H, t\}$ contains three bosons $\{W, Z, H\}$ and one fermion $\{t\}$. To make it F/B -homogeneous, either t must leave $\mathcal{O}_2$ or $\{W, Z, H\}$ must leave. The framework's processor requires the boson sector to be exactly 4 modes (HCSM-13), so $\{W, Z, H\}$ remains in the boson sector together with the dark photon. The dark photon must be joined to $\{W, Z, H\}$ to give a size-4 boson class. Hence t leaves $\mathcal{O}_2$ (to join the fermions) and dark joins $\{W, Z, H\}$ (to complete the boson class). The unique minimal swap is $\{t, \text{dark}\}$. $\square$

Corollary 19 (Orbit-class correspondence). *The winding class partition refines the fermion/boson split: the fermion sector F of the processor is the union $A \cup B$, and the boson sector is C . Equivalently, $T = I - P_C$, as stated and proved in HCSM-13.*

7 Anticommutation with the chiral reflection

7.1 The chiral reflection rs

The diagonal reflection $rs \in D_4$ acts on the mode coordinates by $(n_1, n_2) \mapsto (n_2, n_1)$. On the 14-mode multiplet, its fixed set is

$$\text{Fix}(rs) = \{(2, 2), (6, 6)\} = \{W, t\}, \quad (35)$$

as proved in HCSM-26 [8]. Let P_{rs} be the 14×14 permutation matrix implementing rs on the mode basis.

Theorem 20 (Anticommutation). *The winding generator anticommutes with the chiral reflection:*

$$\{H_{\text{wind}}, P_{rs}\} = H_{\text{wind}}P_{rs} + P_{rs}H_{\text{wind}} = 0. \quad (36)$$

Proof. Let $W = \sin \phi$, so $H_{\text{wind}} = iW$. Then

$$\{H_{\text{wind}}, P_{rs}\} = \{iW, P_{rs}\} = i\{W, P_{rs}\} = i(WP_{rs} + P_{rs}W). \quad (37)$$

It suffices to prove $WP_{rs} = -P_{rs}W$.

The diagonal reflection rs permutes the 14 modes by swapping the two coordinate components. On the winding phase, rs acts by $\phi_{ij} \mapsto \phi_{rs(i)rs(j)}$ where $(rs(i))$ denotes the mode obtained from mode i by the coordinate swap. Since $\phi_{ij} = (\pi/8)(n_1^{(i)}n_2^{(j)} - n_1^{(j)}n_2^{(i)})$, direct computation gives

$$\phi_{rs(i)rs(j)} = \frac{\pi}{8} \left(n_2^{(i)}n_1^{(j)} - n_2^{(j)}n_1^{(i)} \right) = -\phi_{ij}. \quad (38)$$

Hence $\sin(\phi_{rs(i)rs(j)}) = -\sin(\phi_{ij})$, i.e., $W_{rs(i)rs(j)} = -W_{ij}$. Equivalently, $P_{rs}W P_{rs}^{-1} = -W$, i.e., $W P_{rs} = -P_{rs}W$. Therefore $\{H_{\text{wind}}, P_{rs}\} = i(W P_{rs} + P_{rs}W) = 0$. $\square$

Corollary 21 (Chirality of rs). *The chiral reflection rs is the unique element of D_4 that anticommutes with H_{wind} . This is the algebraic input to the Chirality Selection Theorem of HCSM-26.*

8 Inputs and derived

8.1 Derived

The following results are proved in this paper:

1. The winding matrix M has entries in $2\mathbb{Z}$ (Lemma 3).
2. The winding phase ϕ has entries in $(\pi/4)\mathbb{Z}$ (Corollary 4).
3. The winding generator $H_{\text{wind}} = i \sin \phi$ is Hermitian with entries in $\mathbb{Q}(\sqrt{2})$ (Proposition 7).
4. Trace identities: $\text{tr}(H_{\text{wind}}) = \text{tr}(H_{\text{wind}}^3) = 0$, $\text{tr}(H_{\text{wind}}^2) = 80$, $\text{tr}(H_{\text{wind}}^4) = 2176$ (Theorem 8).
5. Characteristic polynomial: $p(\lambda) = \lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32)$ (Theorem 10).
6. Spectrum: $\text{spec}(H_{\text{wind}}) = \{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}$ (Corollary 11).
7. Winding class partition $\{A, B, C\}$ of sizes $(3, 7, 4)$ (Theorem 15).
8. Orbit-class reconstruction: the class partition is the D_4 orbit partition plus the unique two-mode swap $\{t, \text{dark}\}$ (Theorem 18).
9. Anticommutation $\{H_{\text{wind}}, P_{rs}\} = 0$ (Theorem 20).

8.2 Inputs

1. The 14-mode multiplet at $\lambda = 4$ and the mode coordinates (HCSM-04).
2. The D_4 point group action on the multiplet (HCSM-05, HCSM-06).
3. The processor T and the fermion/boson split (HCSM-13).
4. The chiral reflection rs and its fixed set $\{W, t\}$ (HCSM-26).
5. The framework constants $d = 4$, $H = 16$, $N = 64$ (HCSM-01).
6. Newton's identities for power sums and coefficients of a monic polynomial (universal mathematics).

8.3 What is not claimed

We do not claim to derive the antisymmetric bilinear $M = n_1 n_2^T - n_2 n_1^T$ from a more primitive object; it is the unique rank-2 antisymmetric bilinear of the two coordinate vectors, which are the only available structure on the mode labels. We do not claim to derive the elementwise-sine function from first principles; it is the natural choice that produces a Hermitian generator with entries in $\mathbb{Q}(\sqrt{2})$ and spectrum with the small-integer form $\{0, \pm 2\sqrt{2}, \pm 4\sqrt{2}\}$. We do not claim to derive the specific content of the size-3 class A from first principles alone; the split of the 10 fermions into sizes $(3, 7)$ is fixed by the orbit-class reconstruction (Theorem 18).

9 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script is provided in the HCSM companion repository at 12-VERIFICATION/HCSM-14_verification.py. It constructs:

1. The 14 mode coordinates.
2. The coordinate vectors n_1, n_2 .
3. The winding matrix M and phase ϕ .
4. The winding generator $H_{\text{wind}} = i \sin \phi$.
5. The chiral reflection P_{rs} .

and verifies:

1. All entries of M are even integers.
2. H_{wind} is Hermitian.
3. $\text{tr}(H_{\text{wind}}) = 0$, $\text{tr}(H_{\text{wind}}^2) = 80$, $\text{tr}(H_{\text{wind}}^3) = 0$, $\text{tr}(H_{\text{wind}}^4) = 2176$.
4. The characteristic polynomial $p(\lambda) = \lambda^{14} - 40\lambda^{12} + 256\lambda^{10}$.
5. The spectrum $\{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}$.
6. The class partition $\{A, B, C\}$.
7. The class partition is the D_4 orbit partition plus the $\{t, \text{dark}\}$ swap.
8. $\{H_{\text{wind}}, P_{rs}\} = 0$ at machine precision.

Running 12-VERIFICATION/run_all.py reproduces every numerical claim in the HCSM paper stack.

10 Conclusion

This paper has established the winding generator spectrum and class partition of the Hodge Complex Standard Model. The winding generator is

$$H_{\text{wind}} = i \sin\left(\frac{\pi}{8} M\right), \quad M = n_1 n_2^T - n_2 n_1^T. \quad (39)$$

It is the unique Hermitian operator on the 14-mode multiplet determined by the two coordinate vectors of the mode labels. Its characteristic polynomial is

$$p(\lambda) = \lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32), \quad (40)$$

derived from the trace identities $\text{tr}(H_{\text{wind}}^2) = 80$ and $\text{tr}(H_{\text{wind}}^4) = 2176$ by Newton's identities. Its spectrum is

$$\text{spec}(H_{\text{wind}}) = \{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}. \quad (41)$$

The 14 modes partition into three winding classes A, B, C of sizes $(3, 7, 4)$, homogeneous in fermion/boson content. The class partition is the D_4 orbit partition of HCSM-05 modified by the

single two-mode swap $\{t, \text{dark}\}$, uniquely forced by the requirement that each class be fermion/boson homogeneous. The winding generator anticommutes with the chiral reflection rs of D_4 , providing the algebraic input to the Chirality Selection Theorem of HCSM-26.

The winding generator is the fundamental operator on which the loop operator of the framework acts. Its class projectors P_A, P_B, P_C define the three channels through which the processor reads the multiplet, and its spectrum is the spectral skeleton of the framework's chirality and mass content.

References

- [1] S. Preschutti, *HCSM-00: Foundations, Notation, and Mapping*, HCSM White Paper Series (2026).
- [2] S. Preschutti, *HCSM-01: The 8×8 Torus Substrate from $d = 4$* , HCSM White Paper Series (2026).
- [3] S. Preschutti, *HCSM-02: The Hodge Complex on Λ* , HCSM White Paper Series (2026).
- [4] S. Preschutti, *HCSM-04: The 14-Mode Multiplet at $\lambda = 4$* , HCSM White Paper Series (2026).
- [5] S. Preschutti, *HCSM-05: Orbit Decomposition and Inverse Participation*, HCSM White Paper Series (2026).
- [6] S. Preschutti, *HCSM-06: Sector-Coarse Correspondence*, HCSM White Paper Series (2026).
- [7] S. Preschutti, *HCSM-13: The Processor T and the F/B -Class Correspondence*, HCSM White Paper Series (2026).
- [8] S. Preschutti, *HCSM-26: The Observer Pair $\{W, t\}$* , HCSM White Paper Series (2026).
- [9] I. Newton, *Arithmetica Universalis* (1707); see also standard treatments in *Matrix Analysis*, R. A. Horn and C. R. Johnson, Cambridge University Press (2013).
- [10] M. Nakahara, *Geometry, Topology and Physics*, 2nd ed. (CRC Press, 2003).

A Web Crawler Metadata

A.1 Dublin Core

- DC.title: HCSM-14: The Winding Generator Spectrum and Class Partition
- DC.creator: Preschutti, Stanley
- DC.creator.orcid: 0009-0004-5445-1744
- DC.date.issued: 2026-09
- DC.type: Text / Preprint / White Paper
- DC.format: application/pdf; text/x-latex
- DC.language: en
- DC.publisher: Entropia Research Institute

- DC.rights: All rights reserved
- DC.subject: Hodge Complex Standard Model; HCSM; winding generator; winding class partition; Hermitian operator; Newton's identities; D_4 orbits; chirality anticommutation; 8x8 torus
- DC.description: This paper establishes the winding generator $H_{\text{wind}} = i \sin((\pi/8)M)$ on the 14-mode multiplet at $\lambda = 4$. The winding matrix $M = n_1 n_2^T - n_2 n_1^T$ has even-integer entries; the generator is Hermitian with entries in $\mathbb{Q}(\sqrt{2})$. Its characteristic polynomial $\lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32)$ is derived via Newton's identities from the trace identities $\text{tr}(H_{\text{wind}}^2) = 80$ and $\text{tr}(H_{\text{wind}}^4) = 2176$. The spectrum $\{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}$ partitions the 14 modes into three winding classes A, B, C of sizes 3, 7, 4, homogeneous in fermion/boson content. The class partition is the D_4 orbit partition plus the single two-mode swap $\{t, \text{dark}\}$. The winding generator anticommutes with the chiral reflection rs .
- DC.identifier: HCSM-14-2026-09

A.2 Schema.org JSON-LD

```
{
  "@context": "https://schema.org",
  "@type": "ScholarlyArticle",
  "name": "HCSM-14: The Winding Generator Spectrum and Class Partition",
  "author": {
    "@type": "Person",
    "name": "Stanley Prescutti",
    "identifier": "https://orcid.org/0009-0004-5445-1744",
    "affiliation": {"@type": "Organization", "name": "Entropia Research Institute"}
  },
  "datePublished": "2026-09",
  "inLanguage": "en",
  "abstract": "The 14-mode multiplet at lambda=4 admits a unique rank-2 antisymmetric bilinear",
  "keywords": ["Hodge Complex Standard Model", "HCSM", "winding generator", "winding class partition"],
  "license": "All rights reserved",
  "identifier": "HCSM-14-2026-09",
  "isPartOf": {"@type": "Periodical", "name": "HCSM White Paper Series"}
}
```

A.3 BibTeX for Automated Indexing

```
@article{PrescuttiHCSM14,
  author = {Prescutti, Stanley},
  title = {{HCSM-14}: The Winding Generator Spectrum and Class Partition},
  journal = {HCSM White Paper Series},
  year = {2026},
  month = {September},
  note = {ORCID: 0009-0004-5445-1744},
  keywords = {Hodge Complex Standard Model, HCSM, winding generator, winding class partition, Newton's identities,  $D_4$  orbits, chirality anticommutation, 8x8 torus},
  abstract = {The 14-mode multiplet at lambda=4 admits a unique rank-2 antisymmetric
```

bilinear of its coordinate vectors. This paper constructs the winding
 generator $H_{\text{wind}} = i \sin((\pi/8)M)$, proves it is Hermitian with entries
 in $\mathbb{Q}(\sqrt{2})$, derives the characteristic polynomial
 $\lambda^{10} (\lambda^2 - 8)(\lambda^2 - 32)$ via Newton's identities from
 the trace identities $\text{tr}(H_{\text{wind}}^2) = 80$ and $\text{tr}(H_{\text{wind}}^4) = 2176$, and
 proves the spectrum partitions the 14 modes into three winding
 classes A, B, C of sizes 3, 7, 4. The class partition is the D4
 orbit partition plus the single two-mode swap {t, dark}. The winding
 generator anticommutes with the chiral reflection rs .}

}